

MYZ 305E – Artificial Intelligence for Geomatics Eng.

2025-2026 Spring Term Project

Assignment Description:

Within the scope of the term project, students are expected to identify a specific spatially explicit artificial intelligence problem and develop a solution that serves as an example of a Geospatial Artificial Intelligence (GeoAI) application. The work should be prepared as a poster in English. It will be submitted as a PDF on the NINOVA platform.

Requirements:

1. **AI Agent Integration (Core Requirement):** A specific geospatial problem must be solved via GeoAI technologies. Crucially, the methodology **must** involve the development and active use of at least one AI Agent (e.g., an autonomous spatial-reasoning agent, an LLM-based orchestrator with mapping tools, or a multi-agent system for spatial logistics).
2. **Code and Demo Submission:** To verify the functionality of the developed application, students must submit their source code and a **short demo video (maximum 2-3 minutes)** via a GitHub Repository link showing the agent and application in action.
3. At least one geospatial dataset should be used.
4. The data source used in the study should be shared and clearly explained in both the presentation and the poster.
5. The work should be supported by tables and figures (workflow, statistics, accuracy analysis, maps, etc., outputs).
6. The work can be done individually or as a team of two people.

Sample Applications (Adapted for AI Agents):

- Autonomous AI Agents for Disaster Response and Routing
- Multi-Agent Systems for Public Health Monitoring
- AI Agents for Real Estate Market Trend Analysis & Price Prediction
- LLM-Powered GeoChatBots for Spatial Querying
- Autonomous Agents for Smart Mobility/Logistics
- AI Agent-driven Agriculture crop yield optimization

Grading Criteria (Rubric):

- **40% - AI Agent Architecture & Functionality:** Complexity, autonomy, and successful execution of the AI Agent (verified via code and demo).
- **30% - Geospatial Analysis & Problem Solving:** Quality of the geospatial dataset used and how effectively the GeoAI problem was solved.
- **15% - Presentation & Demo Video:** Clarity, structure, and professional delivery of the slide deck and screencast.
- **15% - Poster Design & Narrative:** Visual communication of the methodology, workflow, and results.